PHYDSC401T — Mathematical Physics-III

Unit 4: Theory of Probability, Errors and Tensors

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC401T, Full Marks 100 [ESE 70 + CCA 30] | Unit 4: 12 lectures

Notes 6 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

Index

Unit syllabus (as in the syllabus file)

1. Random variables, Binomial, Poisson and Normal distributions.
2. Central Limit Theorem, Law of Large Numbers.
3. Hypothesis testing.
4. Systematic and random errors. Propagation of errors. Standard and probable error. Least-squares fit.
5. Definition of tensor, covariant and contravariant tensor, fundamental operation with tensors, metric tensor, covariant differentiation and Christoffel symbols.

Top 2 high-mark topics in this unit

1. Binomial, Poisson and Normal distributions, with Central Limit Theorem and Law of Large Numbers — the three named distributions form the whole probability half of the unit and carry standard numerical and proof questions in the Gauhati paper.
2. Tensors: covariant and contravariant tensors, metric tensor, covariant differentiation and Christoffel symbols — the second half of the unit, a definition plus derivation topic that normally carries a long answer, with zero AU coverage.

Contents

1. Random Variables and the Binomial Distribution
2. The Poisson Distribution
3. The Normal Distribution
4. Central Limit Theorem and Law of Large Numbers
5. Hypothesis Testing
6. Systematic and Random Errors
7. Propagation of Errors
8. Standard Error and Probable Error
9. Least-Squares Fit
10. Tensors: Covariant and Contravariant
11. Fundamental Operations with Tensors
12. Metric Tensor, Covariant Differentiation and Christoffel Symbols
13. Questions — Short form, then Long form (no AU questions found for this unit)
14. Sources

1. Random Variables and the Binomial Distribution

Discrete and Continuous Random Variables

A random variable X is a real-valued function defined on the sample space of a random experiment, assigning a numerical value to each possible outcome. Random variables are categorized into two fundamental types based on their state space: discrete and continuous.

A discrete random variable X takes on a finite or countably infinite set of distinct values $\{x_1, x_2, x_3, \dots\}$. The probability distribution of a discrete random variable is specified by its probability mass function (PMF), denoted as $P(X = x)$ or $p(x)$, which gives the probability that X takes the exact value x. The probability mass function must satisfy two basic conditions:

$$p(x_i) \geq 0 \quad \text{for all } i$$

$$\sum_i p(x_i) = 1$$

A continuous random variable X takes values continuously over an interval or union of intervals of real numbers. Because the probability of X taking any single exact value is zero, its probability distribution is described by a probability density function (PDF), denoted as $f(x)$. The probability that X falls within an interval $[a, b]$ is given by the integral of the PDF over that range:

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

The probability density function must satisfy the conditions of non-negativity and normalization over the entire real axis:

$$f(x) \geq 0 \quad \text{for all } x \in \mathbb{R}$$

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

Expectation Value and Variance

The expectation value (or mean) of a random variable, denoted by $E[X]$ or μ, represents the long-term arithmetic average of the variable over repeated trials.

For a discrete random variable with PMF $p(x)$, the expectation value is defined as:

$$E[X] = \mu = \sum_i x_i\, p(x_i)$$

For a continuous random variable with PDF $f(x)$, the expectation value is defined as:

$$E[X] = \mu = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

The variance, denoted by $\mathrm{Var}(X)$ or σ^2, measures the spread or dispersion of the random variable around its mean. It is defined as the expectation value of the squared deviation from the mean:

$$\mathrm{Var}(X) = \sigma^2 = E\left[(X - \mu)^2\right]$$

By expanding the squared term inside the expectation operator, we obtain an alternative and computational form for the variance. Distributing the expectation operator gives:

$$E\left[(X - \mu)^2\right] = E\left[X^2 - 2\mu X + \mu^2\right]$$

Using the linearity of expectation, this becomes:

$$E\left[(X - \mu)^2\right] = E[X^2] - 2\mu E[X] + \mu^2$$

Substituting $E[X] = \mu$ into the expression simplifies the relationship to:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2$$

For a discrete random variable, the second moment $E[X^2]$ is evaluated as:

$$E[X^2] = \sum_i x_i^2\, p(x_i)$$

For a continuous random variable, the second moment is evaluated as:

$$E[X^2] = \int_{-\infty}^{\infty} x^2\, f(x)\, dx$$

Binomial Distribution

A Bernoulli trial is a random experiment that has exactly two mutually exclusive outcomes: success (with probability p) and failure (with probability $q = 1 - p$).

Consider a sequence of n independent Bernoulli trials where the probability of success p remains constant across all trials. Let the random variable X denote the total number of successes obtained in these n trials. The variable X follows a Binomial distribution, written as $X \sim \mathrm{Binomial}(n, p)$.

Derivation of the Probability Mass Function

The number of ways to arrange k successes and $(n - k)$ failures in a sequence of n trials is given by the binomial coefficient:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Because each trial is independent, the probability of any specific sequence containing k successes and $(n - k)$ failures is given by the product of individual probabilities:

$$p^k(1-p)^{n-k}$$

Multiplying the number of distinct arrangements by the probability of a single sequence yields the probability mass function of the Binomial distribution:

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \ldots, n$$

To confirm that this function is normalized, we sum the probabilities over all possible values of k and apply the binomial theorem:

$$\sum_{k=0}^{n} P(X=k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = [p + (1-p)]^n = 1^n = 1$$

Derivation of the Mean

The mean of the Binomial distribution is computed using the expectation formula:

$$E[X] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

Since the term for $k = 0$ evaluates to zero, the sum can begin at $k = 1$:

$$E[X] = \sum_{k=1}^{n} k \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

Canceling k in the numerator with $k!$ in the denominator gives:

$$E[X] = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k}$$

Factoring out n and one factor of p from the summation yields:

$$E[X] = np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)![(n-1)-(k-1)]!} p^{k-1} (1-p)^{(n-1)-(k-1)}$$

Substituting a new summation index $j = k - 1$ changes the limits from $j = 0$ to $j = n - 1$:

$$E[X] = np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j}$$

Applying the binomial theorem to the remaining sum results in $[p + (1-p)]^{n-1} = 1$:

$$E[X] = np$$

Derivation of the Variance

To calculate the variance using $\mathrm{Var}(X) = E[X^2] - (E[X])^2$, we first evaluate the factorial moment $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{n} k(k-1)\binom{n}{k}p^k(1-p)^{n-k}$$

The terms for $k=0$ and $k=1$ vanish, allowing the summation to start at $k=2$:

$$E[X(X-1)] = \sum_{k=2}^{n} k(k-1)\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k}$$

Simplifying the factor $k(k-1)$ with $k!$ leaves $(k-2)!$ in the denominator:

$$E[X(X-1)] = \sum_{k=2}^{n} \frac{n!}{(k-2)!(n-k)!}p^k(1-p)^{n-k}$$

Factoring out $n(n-1)p^2$ from the sum gives:

$$E[X(X-1)] = n(n-1)p^2\sum_{k=2}^{n} \frac{(n-2)!}{(k-2)![(n-2)-(k-2)]!}p^{k-2}(1-p)^{(n-2)-(k-2)}$$

Changing the index of summation to $j = k-2$ yields:

$$E[X(X-1)] = n(n-1)p^2\sum_{j=0}^{n-2} \binom{n-2}{j}p^j(1-p)^{n-2-j}$$

Since the sum equals $[p+(1-p)]^{n-2} = 1$, the factorial moment reduces to:

$$E[X(X-1)] = n(n-1)p^2$$

Expanding this product gives:

$$E[X(X-1)] = n^2p^2 - np^2$$

The second moment $E[X^2]$ is related to $E[X(X-1)]$ by linearity:

$$E[X^2] = E[X(X-1)] + E[X]$$

Substituting the calculated values for $E[X(X-1)]$ and $E[X] = np$:

$$E[X^2] = n^2p^2 - np^2 + np$$

Now, substituting $E[X^2]$ and $(E[X])^2 = (np)^2 = n^2p^2$ into the variance formula:

$$\mathrm{Var}(X) = (n^2p^2 - np^2 + np) - n^2p^2$$

Canceling the n^2p^2 terms produces:

$$\mathrm{Var}(X) = np - np^2$$

Factoring out np yields the standard form of the Binomial variance:

$$\mathrm{Var}(X) = np(1-p)$$

2. The Poisson Distribution

Derivation as the Limit of the Binomial Distribution

The Poisson distribution arises as a limiting case of the Binomial distribution for rare events, where the number of trials n becomes very large ($n \to \infty$) and the probability of success p becomes very small ($p \to 0$), such that the average rate parameter $\lambda = np$ remains constant.

We begin with the Binomial probability mass function:

$$P(X = k) = \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

Substituting $p = \frac{\lambda}{n}$ into the Binomial PMF yields:

$$P(X = k) = \frac{n(n-1)(n-2)\dots(n-k+1)}{k!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

We can split and rearrange the terms into separate factors:

$$P(X = k) = \frac{\lambda^k}{k!} \left[\frac{n}{n} \cdot \frac{n-1}{n} \cdot \frac{n-2}{n} \dots \frac{n-k+1}{n}\right] \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-k}$$

Now we evaluate the limit of each product factor as $n \to \infty$ for a fixed integer k.

First, the product of the rational terms containing n expands as:

$$\lim_{n\to\infty} \left(1 \cdot \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{k-1}{n}\right)\right) = 1$$

Second, using the fundamental limit definition of the exponential function:

$$\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$$

Third, for fixed k, the term raised to the power $-k$ approaches:

$$\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^{-k} = 1^{-k} = 1$$

Combining these three limits gives the probability mass function of the Poisson distribution:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \ldots$$

Probability Mass Function and Normalization

To verify that the Poisson PMF is properly normalized over all non-negative integers:

$$\sum_{k=0}^{\infty} P(X = k) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!}$$

Factoring the constant term $e^{-\lambda}$ outside the summation:

$$\sum_{k=0}^{\infty} P(X = k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

Recognizing the Taylor series expansion $e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$:

$$\sum_{k=0}^{\infty} P(X = k) = e^{-\lambda} e^{\lambda} = e^0 = 1$$

Derivation of Mean and Variance

Derivation of the Mean

The mean of the Poisson distribution is given by:

$$E[X] = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!}$$

The term for $k = 0$ is zero, so the summation index begins at $k = 1$:

$$E[X] = e^{-\lambda} \sum_{k=1}^{\infty} k \frac{\lambda^k}{k!}$$

Canceling k with $k!$ gives:

$$E[X] = e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}$$

Factoring out one power of λ:

$$E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

Shifting the index with $j = k - 1$:

$$E[X] = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Using the series definition for e^{λ}:

$$E[X] = \lambda e^{-\lambda} e^{\lambda} = \lambda$$

Thus, the mean of the Poisson distribution is:

$$E[X] = \lambda$$

Derivation of the Variance

To evaluate the variance, we calculate the factorial moment $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k e^{-\lambda}}{k!}$$

The terms for $k = 0$ and $k = 1$ are zero, so the sum starts at $k = 2$:

$$E[X(X-1)] = e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}$$

Factoring out λ^2:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!}$$

Shifting the index with $j = k - 2$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Substituting the infinite series sum e^{λ}:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2$$

The second moment $E[X^2]$ is:

$$E[X^2] = E[X(X-1)] + E[X] = \lambda^2 + \lambda$$

Substituting $E[X^2]$ and $(E[X])^2 = \lambda^2$ into the variance formula:

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda$$

Thus, both the mean and the variance of the Poisson distribution are equal to λ:

$$E[X] = \mathrm{Var}(X) = \lambda$$

Worked Example: Defective Fuses

Problem Statement: A manufacturing process produces electrical fuses with a 2% defect rate. Fuses are packed in boxes of 200. Find the probability that a box contains almost 5 defective fuses (defined as fewer than 5 defective fuses, i.e., $X < 5$).

Exact Binomial Setup

Let X be the number of defective fuses in a box of $n = 200$. The probability of a fuse being defective is $p = 0.02$. The exact distribution of X is Binomial with PMF:

$$P(X = k) = \binom{200}{k} (0.02)^k (0.98)^{200-k}$$

The exact probability of finding almost 5 defective fuses ($k = 0, 1, 2, 3, 4$) is expressed as:

$$P(X < 5) = \sum_{k=0}^{4} \binom{200}{k} (0.02)^k (0.98)^{200-k}$$

Poisson Approximation

Because $n = 200$ is large and $p = 0.02$ is small, we use the Poisson approximation with parameter λ:

$$\lambda = np = 200 \times 0.02 = 4$$

The Poisson PMF with parameter $\lambda = 4$ is:

$$P(X = k) = \frac{4^k e^{-4}}{k!}$$

The probability of finding fewer than 5 defective fuses is the sum of probabilities for $k = 0, 1, 2, 3, 4$:

$$P(X < 5) = P(X = 0) + P(X = 1) + P(X = 2) + P(X = 3) + P(X = 4)$$

Expanding each term individually using the PMF formula:

$$P(X = 0) = \frac{4^0 e^{-4}}{0!} = e^{-4}$$

$$P(X = 1) = \frac{4^1 e^{-4}}{1!} = 4e^{-4}$$

$$P(X = 2) = \frac{4^2 e^{-4}}{2!} = \frac{16}{2} e^{-4} = 8e^{-4}$$

$$P(X = 3) = \frac{4^3 e^{-4}}{3!} = \frac{64}{6} e^{-4} = \frac{32}{3} e^{-4}$$

$$P(X = 4) = \frac{4^4 e^{-4}}{4!} = \frac{256}{24} e^{-4} = \frac{32}{3} e^{-4}$$

Summing all five terms by factoring out e^{-4}:

$$P(X<5) = e^{-4}\left(1+4+8+\frac{32}{3}+\frac{32}{3}\right)$$

Combining the integer terms:

$$P(X<5) = e^{-4}\left(13+\frac{64}{3}\right)$$

Converting to a single fraction:

$$P(X<5) = e^{-4}\left(\frac{39+64}{3}\right) = \frac{103}{3}e^{-4}$$

Using the numerical value $e^{-4} \approx 0.0183156$:

$$P(X<5) \approx \frac{103}{3} \times 0.0183156 \approx 34.3333 \times 0.0183156 \approx 0.6288$$

Thus, the probability of finding almost 5 defective fuses in a box of 200 is approximately 0.6288 (or 62.88%).

3. The Normal Distribution

Probability Density Function

The Normal (or Gaussian) distribution is a continuous probability distribution whose probability density function $f(x)$ is defined for all real values $x \in (-\infty, \infty)$ by:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

where μ is the location parameter representing the mean, and $\sigma > 0$ is the scale parameter representing the standard deviation.

To prove that $f(x)$ is normalized, we define the integral I:

$$I = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \, dx$$

Substituting $z = \frac{x-\mu}{\sigma}$, so that $dx = \sigma \, dz$:

$$I = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} \, dz$$

Squaring the integral and expressing it as a double integral in Cartesian coordinates (z_1, z_2):

$$I^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(z_1^2+z_2^2)/2}\, dz_1\, dz_2$$

Transforming to polar coordinates (r, θ) with $z_1^2 + z_2^2 = r^2$ and area element $dz_1\, dz_2 = r\, dr\, d\theta$:

$$I^2 = \frac{1}{2\pi} \int_0^{2\pi} d\theta \int_0^{\infty} r e^{-r^2/2}\, dr$$

Evaluating the angular integral:

$$I^2 = \frac{2\pi}{2\pi} \int_0^{\infty} r e^{-r^2/2}\, dr = \left[-e^{-r^2/2}\right]_0^{\infty} = 0 - (-1) = 1$$

Since $I^2 = 1$ and $f(x) > 0$, it follows that $I = 1$, confirming normalization.

Standard Normal Variable

A random variable X following a Normal distribution with mean μ and variance σ^2 is denoted as $X \sim N(\mu, \sigma^2)$.

The Standard Normal Variable Z is defined by the transformation:

$$Z = \frac{X - \mu}{\sigma}$$

The transformation shifts the mean to zero and scales the standard deviation to unity, so $Z \sim N(0, 1)$. The PDF of the standard normal variable, denoted $\phi(z)$, is given by:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

The cumulative distribution function (CDF) of the standard normal distribution is expressed as:

$$\Phi(z) = P(Z \leq z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-u^2/2}\, du$$

Properties of the Normal Distribution

1. **Symmetry:** The probability density function $f(x)$ is symmetric about the mean $x = \mu$. Substituting $(x - \mu) = -(x - \mu)$ yields the exact same value of $f(x)$, meaning $f(\mu - y) = f(\mu + y)$. As a consequence of this symmetry, the mean, median, and mode of the distribution are all identical and located at $x = \mu$.

1. **Maximum Value:** The peak (mode) of the density function occurs at $x = \mu$. Evaluating the PDF at this point yields its maximum value:

$$f(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$$

1. **Points of Inflection:** The points of inflection are the points where the curvature of the PDF curve changes sign, found by setting the second derivative $f''(x) = 0$.

Differentiating $f(x)$ with respect to x:

$$f'(x) = -\frac{x-\mu}{\sigma^2} f(x)$$

Differentiating a second time using the product rule:

$$f''(x) = -\frac{1}{\sigma^2} f(x) - \frac{x-\mu}{\sigma^2} f'(x)$$

Substituting the expression for $f'(x)$ into $f''(x)$:

$$f''(x) = -\frac{1}{\sigma^2} f(x) + \frac{(x-\mu)^2}{\sigma^4} f(x)$$

Factoring out $f(x)$:

$$f''(x) = f(x) \left[\frac{(x-\mu)^2 - \sigma^2}{\sigma^4} \right]$$

Setting $f''(x) = 0$ requires $(x-\mu)^2 - \sigma^2 = 0$, since $f(x) \neq 0$:

$$(x-\mu)^2 = \sigma^2$$

Taking the square root gives the locations of the inflection points:

$$x = \mu \pm \sigma$$

The 68-95-99.7 Rule

The empirical rule (68-95-99.7 rule) specifies the percentage of values that lie within specific standard deviation bounds around the mean for a Normal distribution.

1. **One Standard Deviation ($\mu \pm \sigma$):** The probability that an observation lies within one standard deviation of the mean is:

$$P(\mu - \sigma \leq X \leq \mu + \sigma) = P(-1 \leq Z \leq 1) = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{-z^2/2} \, dz \approx 0.6827$$

This corresponds to approximately 68.27% of the total probability area.

1. **Two Standard Deviations ($\mu \pm 2\sigma$):** The probability that an observation lies within two standard deviations of the mean is:

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) = P(-2 \leq Z \leq 2) = \frac{1}{\sqrt{2\pi}} \int_{-2}^{2} e^{-z^2/2} \, dz \approx 0.9545$$

This corresponds to approximately 95.45% of the total probability area.

1. **Three Standard Deviations ($\mu \pm 3\sigma$):** The probability that an observation lies within three standard deviations of the mean is:

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) = P(-3 \leq Z \leq 3) = \frac{1}{\sqrt{2\pi}} \int_{-3}^{3} e^{-z^2/2}\, dz \approx 0.9973$$

This corresponds to approximately 99.73% of the total probability area, indicating that almost all outcomes of a normal process fall within three standard deviations of the mean.

4. Central Limit Theorem and Law of Large Numbers

The Law of Large Numbers

The Law of Large Numbers describes the long-term stability of the sample mean of a repeated random experiment. Let $X_1, X_2, \ldots, X_N$ be a sequence of N independent and identically distributed (i.i.d.) random variables, each having a common finite expectation value $E[X_i] = \mu$ and a common finite variance $\mathrm{Var}(X_i) = \sigma^2$.

The sample mean of these N random variables, denoted by $\bar{X}_N$, is defined as

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^{N} X_i$$

To understand the behavior of the sample mean, we evaluate its expectation value using the linearity property of the expectation operator:

$$E[\bar{X}_N] = E\left[\frac{1}{N} \sum_{i=1}^{N} X_i\right]$$

Factoring out the constant $1/N$ yields

$$E[\bar{X}_N] = \frac{1}{N} E\left[\sum_{i=1}^{N} X_i\right]$$

Distributing the expectation operator over the sum gives

$$E[\bar{X}_N] = \frac{1}{N} \sum_{i=1}^{N} E[X_i]$$

Substituting $E[X_i] = \mu$ into the summation leads to

$$E[\bar{X}_N] = \frac{1}{N} \sum_{i=1}^{N} \mu$$

Summing the constant μ over N terms yields

$$E[\bar{X}_N] = \frac{1}{N}(N\mu)$$

Simplifying the factor of N results in

$$E[\bar{X}_N] = \mu$$

Thus, the sample mean $\bar{X}_N$ is an unbiased estimator of the population expectation value μ.

We next determine the variance of the sample mean by applying the variance operator:

$$\mathrm{Var}(\bar{X}_N) = \mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} X_i\right)$$

Taking the constant $1/N$ outside the variance operator squarer it:

$$\mathrm{Var}(\bar{X}_N) = \frac{1}{N^2}\mathrm{Var}\left(\sum_{i=1}^{N} X_i\right)$$

Because the random variables $X_1, X_2, \ldots, X_N$ are independent, the variance of the sum is equal to the sum of the variances:

$$\mathrm{Var}(\bar{X}_N) = \frac{1}{N^2}\sum_{i=1}^{N}\mathrm{Var}(X_i)$$

Substituting $\mathrm{Var}(X_i) = \sigma^2$ into the sum gives

$$\mathrm{Var}(\bar{X}_N) = \frac{1}{N^2}\sum_{i=1}^{N}\sigma^2$$

Summing the N variance terms gives

$$\mathrm{Var}(\bar{X}_N) = \frac{1}{N^2}(N\sigma^2)$$

Simplifying the expression yields

$$\mathrm{Var}(\bar{X}_N) = \frac{\sigma^2}{N}$$

To formulate the Weak Law of Large Numbers, we use Chebyshev's inequality. For any random variable Y with mean μ_Y and variance σ_Y^2, Chebyshev's inequality states that for any positive constant $\varepsilon > 0$,

$$P(|Y - \mu_Y| \geq \varepsilon) \leq \frac{\sigma_Y^2}{\varepsilon^2}$$

Setting $Y = \bar{X}_N$, $\mu_Y = \mu$, and $\sigma_Y^2 = \frac{\sigma^2}{N}$, we substitute these quantities into Chebyshev's inequality:

$$P(|\bar{X}_N - \mu| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_N)}{\varepsilon^2}$$

Substituting the expression for $\mathrm{Var}(\bar{X}_N)$ gives

$$P(|\bar{X}_N - \mu| \geq \varepsilon) \leq \frac{\sigma^2}{N\varepsilon^2}$$

Taking the limit as the number of trials N approaches infinity on both sides yields

$$\lim_{N\to\infty} P(|\bar{X}_N - \mu| \geq \varepsilon) \leq \lim_{N\to\infty} \frac{\sigma^2}{N\varepsilon^2}$$

Since σ^2 and ε^2 are fixed constants, the limit on the right-hand side evaluates to zero:

$$\lim_{N\to\infty} \frac{\sigma^2}{N\varepsilon^2} = 0$$

Because probabilities are non-negative, the limit on the left-hand side is bounded between 0 and 0, giving

$$\lim_{N\to\infty} P(|\bar{X}_N - \mu| \geq \varepsilon) = 0$$

The Weak Law of Large Numbers states that as the sample size N grows to infinity, the sample mean $\bar{X}_N$ converges in probability to the true expectation value μ. Mathematically, this means that for any arbitrarily small positive threshold ε, the probability that the deviation $\left|\bar{X}_N - \mu\right|$ exceeds ε vanishes in the limit $N \to \infty$.

The Central Limit Theorem

While the Law of Large Numbers guarantees that the sample mean converges to the expectation value, the Central Limit Theorem (CLT) describes the specific shape of the probability distribution of the sample mean (or total sum) as the sample size becomes large.

Let $X_1, X_2, \ldots, X_N$ be a sequence of independent and identically distributed random variables, each with finite mean μ and finite variance $\sigma^2 > 0$. Consider the sum of these N random variables defined as

$$S_N = \sum_{i=1}^{N} X_i$$

The expectation value of S_N is

$$E[S_N] = E\left[\sum_{i=1}^{N} X_i\right]$$

Using the linearity of expectation, this becomes

$$E[S_N] = \sum_{i=1}^{N} E[X_i] = N\mu$$

The variance of S_N for independent variables is

$$\mathrm{Var}(S_N) = \mathrm{Var}\left[\sum_{i=1}^{N} X_i\right]$$

Summing the individual variances yields

$$\mathrm{Var}(S_N) = \sum_{i=1}^{N} \mathrm{Var}(X_i) = N\sigma^2$$

We define the standardized variable Z_N for the sum S_N by subtracting its mean $N\mu$ and dividing by its standard deviation $\sqrt{N\sigma^2}$:

$$Z_N = \frac{S_N - N\mu}{\sqrt{N\sigma^2}}$$

Dividing both the numerator and denominator by N expresses Z_N directly in terms of the sample mean $\bar{X}_N$:

$$Z_N = \frac{\frac{S_N}{N} - \mu}{\frac{\sqrt{N\sigma^2}}{N}}$$

Simplifying the algebraic terms gives

$$Z_N = \frac{\bar{X}_N - \mu}{\sigma/\sqrt{N}}$$

The Central Limit Theorem states that as $N \to \infty$, the cumulative distribution function of Z_N converges point-wise to the cumulative distribution function of the standard normal distribution $N(0,1)$. That is, for any real numbers $a < b$,

$$\lim_{N\to\infty} P(a \leq Z_N \leq b) = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-z^2/2}\,dz$$

This limit holds regardless of the underlying probability distribution of the individual variables X_i (whether continuous, discrete, symmetric, or skewed), provided the mean and variance are finite. For a finite, large N, the sample mean $\bar{X}_N$ is approximately normally distributed:

$$\bar{X}_N \sim N\left(\mu, \frac{\sigma^2}{N}\right)$$

Similarly, the sum S_N is approximately normally distributed:

$$S_N \sim N\left(N\mu, N\sigma^2\right)$$

Significance of the Central Limit Theorem in Measurement Errors

In physical experiments, an observed measurement value X_{obs} is composed of the true value x_{true} and a net measurement error ϵ:

$$X_{\text{obs}} = x_{\text{true}} + \epsilon$$

In experimental setups, the overall error ϵ is rarely caused by a single isolated source. Instead, ϵ represents the combined sum of a large number N of microscopic, independent random disturbances δ_i:

$$\epsilon = \sum_{i=1}^{N} \delta_i$$

These individual disturbances originate from distinct physical effects, such as thermal fluctuations in electronic components, small mechanical vibrations, voltage instabilities, background radiation noise, and environmental temperature shifts. Even if each individual disturbance δ_i has a non-normal distribution (such as a uniform, triangular, or asymmetric distribution), the Central Limit Theorem proves that their sum $\epsilon = \sum_{i=1}^{N} \delta_i$ tends toward a Gaussian (normal) distribution as the number of small noise sources N becomes large.

This result justifies the standard practice in physical sciences of assuming that random experimental measurement errors follow a normal distribution centered at zero mean, enabling the application of standard Gaussian error analysis techniques.

5. Hypothesis Testing

Null and Alternative Hypotheses

Statistical hypothesis testing provides a framework for determining whether experimental data support a theoretical prediction or physical model.

The hypothesis testing process begins by formulating two competing statements:

1. **Null Hypothesis (H_0):** The default statement assuming no physical effect, no signal, no difference, or that a parameter equals a specific theoretical baseline value μ_0. It is written as

$$H_0 : \mu = \mu_0$$

1. **Alternative Hypothesis (H_1 or H_a):** The claim that replaces H_0 if the experimental evidence against H_0 is sufficiently strong. It asserts the presence of a real physical effect, shift, or difference. The alternative hypothesis can be two-sided:

$$H_1 : \mu \neq \mu_0$$

or one-sided (directional):

$$H_1 : \mu > \mu_0 \quad \text{or} \quad H_1 : \mu < \mu_0$$

Test Statistic and Decision Rule

To test H_0, a sample of size N is collected from an experiment, and a test statistic is calculated.

The test statistic is a single random variable calculated from sample observations whose probability distribution under H_0 is known. For example, when testing the mean μ of a normally distributed quantity with known standard deviation σ, the standard test statistic Z is computed as

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{N}}$$

where $\bar{X}$ is the sample mean observed in the experiment. Under H_0, Z follows the standard normal distribution $N(0, 1)$.

The **significance level** (α) is a pre-chosen probability threshold that defines the maximum allowed probability of rejecting H_0 when H_0 is true. Common choices in physics and data analysis are $\alpha = 0.05$, $\alpha = 0.01$, or tighter thresholds such as 3σ ($\alpha \approx 0.0027$) and 5σ ($\alpha \approx 3 \times 10^{-7}$).

The **decision rule** divides the set of all possible values of the test statistic into an acceptance region and a critical region (rejection region C).

For a two-tailed test at significance level α, the critical region consists of both extreme tails of the standard normal distribution. The critical value $z_{\alpha/2}$ satisfies

$$P(Z > z_{\alpha/2}) = \frac{\alpha}{2}$$

The decision rule is formulated as:

$$\text{Reject } H_0 \quad \text{if } |Z| \geq z_{\alpha/2}$$

$$\text{Fail to reject } H_0 \quad \text{if } |Z| < z_{\alpha/2}$$

For a one-tailed upper-tail test at significance level α, the critical value z_α satisfies

$$P(Z > z_\alpha) = \alpha$$

The decision rule is:

$$\text{Reject } H_0 \quad \text{if } Z \geq z_\alpha$$

$$\text{Fail to reject } H_0 \quad \text{if } Z < z_\alpha$$

Type I and Type II Errors

Because hypothesis testing is based on random sample data, statistical decisions carry the risk of making an error. Two distinct types of errors can occur:

1. **Type I Error:** Occurs when the null hypothesis H_0 is true, but the decision rule leads to the rejection of H_0 (false positive). The probability of committing a Type I error is equal to the significance level α:

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ is true})$$

1. **Type II Error:** Occurs when the null hypothesis H_0 is false, but the decision rule fails to reject H_0 (false negative). The probability of committing a Type II error is denoted by β:

$$\beta = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$$

The **Power of a Test** is defined as the probability of correctly rejecting H_0 when H_0 is false (i.e., when H_1 is true). It is given by

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_1 \text{ is true})$$

The decisions and possible error types are summarized below:

Decision	**Null Hypothesis H_0 is True**	**Null Hypothesis H_0 is False (H_1 is True)**
Fail to Reject H_0	Correct Decision ($1 - \alpha$)	Type II Error (β)
Reject H_0	Type I Error (α)	Correct Decision / Power ($1 - \beta$)

The p-Value

The p-value provides a quantitative measure of consistency between the observed sample data and the null hypothesis.

The p-value is defined as the probability, calculated under the assumption that H_0 is strictly true, of obtaining a test statistic value equal to or more extreme than the observed value z_{obs}.

For a two-tailed test with a standard normal test statistic Z, the p-value is calculated as

$$p\text{-value} = P(|Z| \geq |z_{\text{obs}}| \mid H_0 \text{ is true})$$

Evaluating this probability using the standard normal cumulative distribution function $\Phi(z)$ gives

$$p\text{-value} = 2\left[1 - \Phi(|z_{\text{obs}}|)\right]$$

For a right-tailed test, the p-value is

$$p\text{-value} = P(Z \geq z_{\text{obs}} \mid H_0 \text{ is true}) = 1 - \Phi(z_{\text{obs}})$$

The p-value is directly linked to the decision rule: - If $p\text{-value} \leq \alpha$, the test statistic falls in the critical region, and H_0 is rejected at significance level α. - If $p\text{-value} > \alpha$, the test statistic falls outside the critical region, and H_0 cannot be rejected at significance level α.

Clarification of what the p-value IS and what it IS NOT:

What the p-value IS: - It IS the probability of obtaining data as extreme as, or more extreme than, the observed data, assuming H_0 is true. - It IS a continuous measure of statistical evidence against H_0; smaller p-values indicate stronger incompatibility between the observed data and H_0.

What the p-value IS NOT: - It IS NOT the probability that the null hypothesis is true, $P(H_0 \mid \text{data})$. - It IS NOT the probability that the alternative hypothesis is false, $P(H_1 \text{ is false} \mid \text{data})$. - It IS NOT the probability that the experimental result was produced by random chance alone. - It IS NOT the probability of making a Type I error (the Type I error probability is fixed in advance by α, whereas the p-value is a random variable that varies from sample to sample).

6. Systematic and Random Errors

Definitions and Differences

In experimental physics, every measurement of a physical quantity is subject to experimental uncertainties. The difference between the observed value x_{obs} and the true value x_{true} of a quantity is defined as the experimental error:

$$\text{Error} = x_{\text{obs}} - x_{\text{true}}$$

Experimental errors are fundamentally classified into two categories: systematic errors and random errors.

Systematic errors are reproducible inaccuracies that consistently bias measurements in one direction, causing observed values to be either systematically higher or systematically lower than the true value. They arise from identifiable flaws in equipment, calibration, or experimental procedure.

Random errors are unpredictable, non-reproducible variations caused by unobserved or uncontrollable fluctuations in environmental conditions, instrument limitations, or human judgment. They cause measured values to scatter symmetrically about the mean value.

The essential differences between systematic and random errors are summarized below:

1. Directionality: Systematic errors introduce a definite bias in a single direction (positive or negative). Random errors fluctuate in both positive and negative directions with equal likelihood.
2. Predictability: Systematic errors remain constant or vary deterministically under identical experimental conditions. Random errors follow a statistical distribution, typically a Gaussian (Normal) distribution.
3. Detection: Systematic errors cannot be detected by simply repeating measurements under identical conditions, as the same error recurs each time. Random errors are readily revealed through the scatter of repeated observations.

Examples in Physics Experiments

Examples of systematic errors include:

1. Zero Error of Instruments: A vernier caliper whose zero mark does not coincide with the main scale zero mark when the jaws are closed introduces a constant offset to every length measurement.
2. Improper Instrument Calibration: A voltmeter calibrated incorrectly at a higher temperature than the operational environment gives readings scaled by a constant factor.

3. Environmental Drift: Thermal expansion of a metal measuring tape during an experiment performed at elevated temperatures causes all measured distances to be systematically underestimated.

Examples of random errors include:

1. Parallax Error in Visual Readings: Small variations in eye placement when reading an analog meter introduce small, random fluctuations above and below the true reading.
2. Environmental Vibrations: Mechanical vibrations in a laboratory floor cause minor, random variations in the equilibrium position of a sensitive balance beam.
3. Electronic Noise: Thermal noise within amplifier circuits causes random fluctuations in voltage readings taken across a resistor.

Reduction of Errors through Repetition

Random errors obey the laws of probability. Consider N independent measurements $x_1, x_2, \ldots, x_N$ of a quantity x, each having an independent random deviation ϵ_i with zero mean and variance σ^2. The mean of the measurements is given by:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^{N} x_i$$

The variance of the mean value is:

$$\sigma_{\bar{x}}^2 = \mathrm{Var}\left(\frac{1}{N} \sum_{i=1}^{N} x_i \right)$$

Since individual measurements are statistically independent, the variance of the sum equals the sum of the variances:

$$\sigma_{\bar{x}}^2 = \frac{1}{N^2} \sum_{i=1}^{N} \mathrm{Var}(x_i)$$

Substituting $\mathrm{Var}(x_i) = \sigma^2$ into the sum yields:

$$\sigma_{\bar{x}}^2 = \frac{1}{N^2} \cdot N\sigma^2 = \frac{\sigma^2}{N}$$

Taking the square root gives the standard deviation of the mean:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}}$$

As the number of measurements N increases, the standard deviation of the mean decreases proportional to $1/\sqrt{N}$. Thus, random errors are reduced by repeated observations and averaging.

In contrast, systematic errors are constant biases present in every trial. Let c be a constant systematic offset such that each measurement yields $x_i = x_{\text{true}} + c + \epsilon_i$. Taking the mean of N measurements yields:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N}(x_{\text{true}} + c + \epsilon_i)$$

Splitting the sum into separate terms gives:

$$\bar{x} = x_{\text{true}} + c + \frac{1}{N}\sum_{i=1}^{N}\epsilon_i$$

Taking the expectation value of both sides, noting that $\langle \epsilon_i \rangle = 0$, results in:

$$\langle \bar{x} \rangle = x_{\text{true}} + c$$

The systematic bias c remains entirely unchanged regardless of how large N becomes. Repetition cannot reduce or eliminate systematic errors; they can only be eliminated by careful instrument calibration, proper experimental design, or applying known analytical corrections.

7. Propagation of Errors

General Formula for Error Propagation

When a physical quantity z cannot be measured directly but is calculated from several independent measured quantities $x_1, x_2, \ldots, x_n$ via a function $z = f(x_1, x_2, \ldots, x_n)$, the uncertainties in x_i propagate into the calculated value z.

Let $\bar{x}_i$ denote the mean value of variable x_i, and let $\Delta x_i = x_i - \bar{x}_i$ be the small deviation of x_i from its mean value. Expanding $f(x_1, x_2, \ldots, x_n)$ in a multivariable Taylor series about the mean values up to first-order terms yields:

$$f(x_1, x_2, \ldots, x_n) \approx f(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n) + \sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}\right)(x_i - \bar{x}_i)$$

Subtracting $f(\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_n) = \bar{z}$ from both sides gives the deviation in z:

$$\Delta z = z - \bar{z} = \sum_{i=1}^{n}\left(\frac{\partial f}{\partial x_i}\right)\Delta x_i$$

Squaring both sides of this expression yields:

$$(\Delta z)^2 = \left(\sum_{i=1}^{n}\frac{\partial f}{\partial x_i}\Delta x_i\right)\left(\sum_{j=1}^{n}\frac{\partial f}{\partial x_j}\Delta x_j\right)$$

Expanding the double sum explicitly:

$$(\Delta z)^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 (\Delta x_i)^2 + \sum_{i \neq j} \left(\frac{\partial f}{\partial x_i} \right) \left(\frac{\partial f}{\partial x_j} \right) \Delta x_i \Delta x_j$$

Taking the expectation value (ensemble average) across all measurements gives:

$$\langle (\Delta z)^2 \rangle = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 \langle (\Delta x_i)^2 \rangle + \sum_{i \neq j} \left(\frac{\partial f}{\partial x_i} \right) \left(\frac{\partial f}{\partial x_j} \right) \langle \Delta x_i \Delta x_j \rangle$$

Assuming the measurements of variables x_i and x_j are completely independent and uncorrelated, the covariance term vanishes: $\langle \Delta x_i \Delta x_j \rangle = 0$ for $i \neq j$. Defining the variance $\sigma_z^2 = \langle (\Delta z)^2 \rangle$ and $\sigma_{x_i}^2 = \langle (\Delta x_i)^2 \rangle$, the general equation for error propagation in quadrature is obtained:

$$\sigma_z^2 = \sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2$$

Taking the square root of both sides gives the standard error formula:

$$\sigma_z = \sqrt{\sum_{i=1}^{n} \left(\frac{\partial f}{\partial x_i} \right)^2 \sigma_{x_i}^2}$$

Error in a Sum

Consider a function representing the sum of two independent measured variables x and y:

$$z = x + y$$

The partial derivative of z with respect to x is:

$$\frac{\partial z}{\partial x} = 1$$

The partial derivative of z with respect to y is:

$$\frac{\partial z}{\partial y} = 1$$

Substituting these partial derivatives into the general error propagation formula gives:

$$\sigma_z^2 = \left(\frac{\partial z}{\partial x} \right)^2 \sigma_x^2 + \left(\frac{\partial z}{\partial y} \right)^2 \sigma_y^2$$

Substituting the evaluated derivatives into the equation yields:

$$\sigma_z^2 = (1)^2 \sigma_x^2 + (1)^2 \sigma_y^2$$

$$\sigma_z^2 = \sigma_x^2 + \sigma_y^2$$

Taking the square root yields the absolute uncertainty in a sum:

$$\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$$

Worked Numerical Example for a Sum

Suppose the lengths of two separate metal rods are measured independently as $x = 12.4\ \text{cm}$ with an uncertainty of $\sigma_x = 0.3\ \text{cm}$, and $y = 8.2\ \text{cm}$ with an uncertainty of $\sigma_y = 0.4\ \text{cm}$. Calculate the total combined length $z = x + y$ and its propagated error σ_z.

Calculating the mean total length:

$$z = 12.4\ \text{cm} + 8.2\ \text{cm} = 20.6\ \text{cm}$$

Applying the formula for the error in a sum:

$$\sigma_z = \sqrt{\sigma_x^2 + \sigma_y^2}$$

Substituting the given values into the formula:

$$\sigma_z = \sqrt{(0.3)^2 + (0.4)^2}$$

$$\sigma_z = \sqrt{0.09 + 0.16}$$

$$\sigma_z = \sqrt{0.25} = 0.5\ \text{cm}$$

Thus, the final result is reported as $z = 20.6 \pm 0.5\ \text{cm}$.

Error in a Product

Consider a function representing the product of two independent measured variables x and y:

$$z = xy$$

The partial derivative of z with respect to x is:

$$\frac{\partial z}{\partial x} = y$$

The partial derivative of z with respect to y is:

$$\frac{\partial z}{\partial y} = x$$

Substituting these derivatives into the general error propagation equation:

$$\sigma_z^2 = \left(\frac{\partial z}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial z}{\partial y}\right)^2 \sigma_y^2$$

Substituting y and x into the expression:

$$\sigma_z^2 = y^2\sigma_x^2 + x^2\sigma_y^2$$

Dividing both sides of this equation by $z^2 = x^2y^2$:

$$\frac{\sigma_z^2}{z^2} = \frac{y^2\sigma_x^2 + x^2\sigma_y^2}{x^2y^2}$$

Simplifying each term on the right-hand side:

$$\frac{\sigma_z^2}{z^2} = \frac{y^2\sigma_x^2}{x^2y^2} + \frac{x^2\sigma_y^2}{x^2y^2}$$

$$\frac{\sigma_z^2}{z^2} = \left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2$$

Taking the square root gives the relative uncertainty in a product:

$$\frac{\sigma_z}{z} = \sqrt{\left(\frac{\sigma_x}{x}\right)^2 + \left(\frac{\sigma_y}{y}\right)^2}$$

Worked Numerical Example for a Product

A constant force $F = 20.0\ \text{N}$ with an absolute uncertainty $\sigma_F = 0.5\ \text{N}$ acts on an object causing it to move over a distance $d = 4.0\ \text{m}$ with an absolute uncertainty $\sigma_d = 0.2\ \text{m}$. Calculate the work done $W = F \cdot d$ and its uncertainty σ_W.

Calculating the mean value of work done:

$$W = F \cdot d = 20.0\ \text{N} \times 4.0\ \text{m} = 80.0\ \text{J}$$

Calculating the relative uncertainty squared for F:

$$\left(\frac{\sigma_F}{F}\right)^2 = \left(\frac{0.5}{20.0}\right)^2 = (0.025)^2 = 0.000625$$

Calculating the relative uncertainty squared for d:

$$\left(\frac{\sigma_d}{d}\right)^2 = \left(\frac{0.2}{4.0}\right)^2 = (0.05)^2 = 0.002500$$

Adding the relative uncertainties squared:

$$\frac{\sigma_W^2}{W^2} = 0.000625 + 0.002500 = 0.003125$$

Taking the square root to find the fractional relative error:

$$\frac{\sigma_W}{W} = \sqrt{0.003125} \approx 0.0559$$

Multiplying by W to determine the absolute error σ_W:

$$\sigma_W = 80.0 \times 0.05590 = 4.472 \text{ J} \approx 4.5 \text{ J}$$

Thus, the calculated work is reported as $W = 80.0 \pm 4.5$ J.

8. Standard Error and Probable Error

Standard Error of a Measurement and Mean

When a quantity is measured N times yielding values $x_1, x_2, \ldots, x_N$, the best estimate of the true value is the sample mean:

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i$$

The variance of the sample measurement distribution, representing the precision of a single measurement, is given by the sample variance s^2 (with $N-1$ degrees of freedom):

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2$$

The standard deviation of a single measurement is $\sigma = \sqrt{s^2}$.

The standard error of the mean (often denoted $\sigma_{\bar{x}}$) quantifies the statistical reliability of the sample mean $\bar{x}$ as an estimator of the true population mean. It is derived using the general principle of error propagation applied to the definition of $\bar{x}$.

Writing the mean explicitly as a function of independent variables:

$$\bar{x}\left(x_1, x_2, \ldots, x_N\right) = \frac{1}{N}x_1 + \frac{1}{N}x_2 + \cdots + \frac{1}{N}x_N$$

Evaluating the partial derivative with respect to any individual observation x_i:

$$\frac{\partial \bar{x}}{\partial x_i} = \frac{1}{N}$$

Using the error propagation formula for independent measurements with identical standard deviation σ:

$$\sigma_{\bar{x}}^2 = \sum_{i=1}^{N}\left(\frac{\partial \bar{x}}{\partial x_i}\right)^2 \sigma_{x_i}^2$$

Substituting $\frac{\partial \bar{x}}{\partial x_i} = \frac{1}{N}$ and $\sigma_{x_i} = \sigma$ into the sum:

$$\sigma_{\bar{x}}^2 = \sum_{i=1}^{N} \left(\frac{1}{N}\right)^2 \sigma^2$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{N^2} \sum_{i=1}^{N} 1$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{N^2} \cdot N = \frac{\sigma^2}{N}$$

Taking the square root of both sides gives the standard error of the mean:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{N}}$$

This relation demonstrates that the uncertainty in the mean value decreases inversely with the square root of the total number of measurements.

Probable Error

The probable error ($P.E.$) is defined as the magnitude of error such that the probability of an experimental error falling within the interval $[-P.E., +P.E.]$ is exactly 0.5 (50%). Equally, there is a 50% chance that the error in a measurement exceeds the probable error in absolute magnitude.

Assuming the experimental errors are distributed normally according to a continuous Gaussian probability density function with mean zero and standard deviation σ:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

By definition of probable error $r = P.E.$, the integral of $p(x)$ over $[-r, r]$ must equal 0.5:

$$\int_{-r}^{r} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}\, dx = 0.5$$

Substituting the dimensionless variable $t = \frac{x}{\sigma}$, with $dx = \sigma\, dt$:

$$\frac{1}{\sqrt{2\pi}} \int_{-r/\sigma}^{r/\sigma} e^{-\frac{t^2}{2}}\, dt = 0.5$$

Due to the symmetry of the integrand about $t = 0$:

$$\frac{2}{\sqrt{2\pi}} \int_{0}^{r/\sigma} e^{-\frac{t^2}{2}}\, dt = 0.5$$

$$\sqrt{\frac{2}{\pi}} \int_{0}^{r/\sigma} e^{-\frac{t^2}{2}}\, dt = 0.5$$

Evaluating this standard normal distribution integral numerically:

$$\frac{r}{\sigma} \approx 0.67449$$

Rounding to four decimal places, the relationship between probable error and standard deviation is:

$$P.E. = 0.6745\,\sigma$$

For the mean of N independent observations, the probable error of the mean ($P.E._{\bar{x}}$) is directly related to the standard error of the mean:

$$P.E._{\bar{x}} = 0.6745\,\sigma_{\bar{x}} = 0.6745\frac{\sigma}{\sqrt{N}}$$

9. Least-Squares Fit

Linear Model and Principle of Least Squares

In experimental physics, it is frequently necessary to determine a linear functional relationship between an independent variable x and a dependent variable y. Given a set of N experimental data points $(x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$, we fit a straight line given by:

$$y = a + bx$$

where a is the y-intercept and b is the slope of the line.

For each data point (x_i, y_i), the vertical residual d_i (the vertical distance between the experimental value y_i and the value predicted by the fitted line) is defined as:

$$d_i = y_i - (a + bx_i)$$

The Principle of Least Squares states that the best-fit line is obtained by choosing parameters a and b such that the sum of the squares of the residuals $S(a, b)$ is minimized:

$$S(a, b) = \sum_{i=1}^{N} d_i^2 = \sum_{i=1}^{N} (y_i - a - bx_i)^2$$

Derivation of Normal Equations

To determine the values of a and b that minimize $S(a, b)$, we take the partial derivatives of S with respect to a and b independently and set each derivative to zero.

Taking the partial derivative of $S(a, b)$ with respect to a:

$$\frac{\partial S}{\partial a} = \frac{\partial}{\partial a} \sum_{i=1}^{N} (y_i - a - bx_i)^2$$

Differentiating term-by-term using the chain rule:

$$\frac{\partial S}{\partial a} = \sum_{i=1}^{N} 2(y_i - a - bx_i)(-1) = -2\sum_{i=1}^{N}(y_i - a - bx_i)$$

Setting $\frac{\partial S}{\partial a} = 0$:

$$-2\sum_{i=1}^{N}(y_i - a - bx_i) = 0$$

Dividing both sides by -2:

$$\sum_{i=1}^{N}(y_i - a - bx_i) = 0$$

Splitting the summation into individual terms:

$$\sum_{i=1}^{N} y_i - \sum_{i=1}^{N} a - b\sum_{i=1}^{N} x_i = 0$$

Since a is a constant, $\sum_{i=1}^{N} a = Na$:

$$\sum_{i=1}^{N} y_i - Na - b\sum_{i=1}^{N} x_i = 0$$

Rearranging terms yields the First Normal Equation:

$$aN + b\sum_{i=1}^{N} x_i = \sum_{i=1}^{N} y_i$$

Next, taking the partial derivative of $S(a, b)$ with respect to b:

$$\frac{\partial S}{\partial b} = \frac{\partial}{\partial b}\sum_{i=1}^{N}(y_i - a - bx_i)^2$$

Applying the chain rule term-by-term:

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{N} 2(y_i - a - bx_i)(-x_i) = -2\sum_{i=1}^{N} x_i(y_i - a - bx_i)$$

Setting $\frac{\partial S}{\partial b} = 0$:

$$-2\sum_{i=1}^{N} x_i(y_i - a - bx_i) = 0$$

Dividing both sides by -2:

$$\sum_{i=1}^{N}(x_i y_i - a x_i - b x_i^2) = 0$$

Splitting the summation into individual terms:

$$\sum_{i=1}^{N} x_i y_i - a\sum_{i=1}^{N} x_i - b\sum_{i=1}^{N} x_i^2 = 0$$

Rearranging terms yields the Second Normal Equation:

$$a\sum_{i=1}^{N} x_i + b\sum_{i=1}^{N} x_i^2 = \sum_{i=1}^{N} x_i y_i$$

The two normal equations form a system of two simultaneous linear equations in two unknowns (a and b):

$$\begin{aligned} 1.\quad & aN + b\sum x_i = \sum y_i \\ 2.\quad & a\sum x_i + b\sum x_i^2 = \sum x_i y_i \end{aligned}$$

To solve for slope b, multiply the first normal equation by $\sum x_i$:

$$aN\sum x_i + b\left(\sum x_i\right)^2 = \left(\sum y_i\right)\left(\sum x_i\right)$$

Multiply the second normal equation by N:

$$aN\sum x_i + bN\sum x_i^2 = N\sum x_i y_i$$

Subtracting the upper equation from the lower equation eliminates a:

$$b\left[N\sum x_i^2 - \left(\sum x_i\right)^2\right] = N\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)$$

Solving explicitly for b:

$$b = \frac{N\sum x_i y_i - \left(\sum x_i\right)\left(\sum y_i\right)}{N\sum x_i^2 - \left(\sum x_i\right)^2}$$

Solving the first normal equation for intercept a:

$$aN = \sum y_i - b\sum x_i$$

Dividing through by N:

$$a = \frac{\sum y_i - b\sum x_i}{N} = \bar{y} - b\bar{x}$$

where $\bar{x} = \frac{1}{N}\sum x_i$ and $\bar{y} = \frac{1}{N}\sum y_i$.

Worked Example on a Small Data Set

Fit a straight line $y = a + bx$ to the following set of $N = 4$ experimental data points using the method of least squares:

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 3), \quad (x_3, y_3) = (3, 5), \quad (x_4, y_4) = (4, 6)$$

Constructing a summation table for the required quantities:

i	x_i	y_i	x_i^2	$x_i y_i$
1	1	2	1	2
2	2	3	4	6
3	3	5	9	15
4	4	6	16	24
$\sum$	**10**	**16**	**30**	**47**

Extracting the scalar summations:

$$N = 4$$

$$\sum x_i = 10$$

$$\sum y_i = 16$$

$$\sum x_i^2 = 30$$

$$\sum x_i y_i = 47$$

Calculating the means:

$$\bar{x} = \frac{\sum x_i}{N} = \frac{10}{4} = 2.5$$

$$\bar{y} = \frac{\sum y_i}{N} = \frac{16}{4} = 4.0$$

Calculating the denominator term for b:

$$\text{Denominator} = N \sum x_i^2 - \left(\sum x_i \right)^2 = 4(30) - (10)^2 = 120 - 100 = 20$$

Calculating the numerator term for b:

$$\text{Numerator} = N \sum x_i y_i - \left(\sum x_i \right) \left(\sum y_i \right) = 4(47) - (10)(16) = 188 - 160 = 28$$

Calculating the slope b:

$$b = \frac{28}{20} = 1.4$$

Calculating the intercept a:

$$a = \bar{y} - b\bar{x} = 4.0 - (1.4)(2.5) = 4.0 - 3.5 = 0.5$$

The equation of the least-squares best-fit straight line is therefore:

$$y = 0.5 + 1.4x$$

10. Tensors: Covariant and Contravariant

Physical laws must maintain their mathematical form under changes of coordinate systems. Tensors are geometric objects defined by specific linear transformation laws under coordinate transformations.

Consider an n-dimensional space governed by a real coordinate system $x^i = (x^1, x^2, \ldots, x^n)$. Under an invertible transformation to a new coordinate system $\bar{x}^i = (\bar{x}^1, \bar{x}^2, \ldots, \bar{x}^n)$, the differentials of the new coordinates are related to the differentials of the old coordinates by the chain rule:

$$d\bar{x}^i = \sum_{j=1}^{n} \frac{\partial \bar{x}^i}{\partial x^j} dx^j$$

Adopting the Einstein summation convention, where any repeated index appearing once as a superscript and once as a subscript implies summation over its full range, this transformation is written as:

$$d\bar{x}^i = \frac{\partial \bar{x}^i}{\partial x^j} dx^j$$

The total number of indices required to specify a tensor component defines the rank (or order) of the tensor.

Scalar (Tensor of Rank 0)

A scalar or invariant is a physical quantity described by a single component that does not change under coordinate transformations. If $\phi(x)$ is a scalar field in the unbarred system and $\bar{\phi}(\bar{x})$ is the same quantity in the barred system, the transformation rule is:

$$\bar{\phi}(\bar{x}) = \phi(x)$$

Vectors (Tensors of Rank 1)

Vectors are rank-1 tensors classified into two distinct operational types based on how their components transform under coordinate transformations.

Contravariant Vectors

A set of components A^i ($i = 1, 2, \ldots, n$) forms a contravariant vector if it transforms in the same manner as the coordinate differentials dx^i. Representing components with upper indices, the transformation law under the

coordinate change $x \to \bar{x}$ is:

$$\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^j} A^j$$

To show that coordinate differentials dx^i constitute a contravariant vector, consider the total differential of $\bar{x}^i$:

$$d\bar{x}^i = \frac{\partial \bar{x}^i}{\partial x^j} dx^j$$

Comparing this directly with the definition of a contravariant vector confirms that dx^i transforms as a contravariant vector of rank 1.

Covariant Vectors

A set of components B_i ($i = 1, 2, \ldots, n$) forms a covariant vector if it transforms with the inverse transformation matrix, represented using lower indices. The transformation law is:

$$\bar{B}_i = \frac{\partial x^j}{\partial \bar{x}^i} B_j$$

To illustrate a covariant vector, consider the gradient of a scalar field $\phi(x)$. Let the components be $B_i = \frac{\partial \phi}{\partial x^i}$. Applying the chain rule to the derivative in the barred coordinate system gives:

$$\frac{\partial \bar{\phi}}{\partial \bar{x}^i} = \frac{\partial \phi}{\partial x^j} \frac{\partial x^j}{\partial \bar{x}^i}$$

Substituting $B_j = \frac{\partial \phi}{\partial x^j}$ and $\bar{B}_i = \frac{\partial \bar{\phi}}{\partial \bar{x}^i}$ into this relation yields:

$$\bar{B}_i = \frac{\partial x^j}{\partial \bar{x}^i} B_j$$

This demonstrates that the gradient of a scalar field transforms as a covariant vector of rank 1.

Higher-Rank Tensors

Tensors of higher rank possess multiple indices, each transforming independently as either a contravariant (upper) or covariant (lower) index.

A contravariant tensor of rank 2 possesses two upper indices and transforms according to:

$$\bar{T}^{ij} = \frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial \bar{x}^j}{\partial x^l} T^{kl}$$

A covariant tensor of rank 2 possesses two lower indices and transforms according to:

$$\bar{T}_{ij} = \frac{\partial x^k}{\partial \bar{x}^i} \frac{\partial x^l}{\partial \bar{x}^j} T_{kl}$$

A mixed tensor of rank 2 has one contravariant (upper) index and one covariant (lower) index, transforming as:

$$\bar{T}^i_j = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial x^l}{\partial \bar{x}^j}T^k_l$$

In general, a mixed tensor $T^{i_1 i_2 \ldots i_r}_{j_1 j_2 \ldots j_s}$ of contravariant rank r and covariant rank s (total rank $r+s$) transforms under a change of coordinates as:

$$\bar{T}^{i_1 i_2 \ldots i_r}_{j_1 j_2 \ldots j_s} = \frac{\partial \bar{x}^{i_1}}{\partial x^{k_1}} \cdots \frac{\partial \bar{x}^{i_r}}{\partial x^{k_r}} \frac{\partial x^{l_1}}{\partial \bar{x}^{j_1}} \cdots \frac{\partial x^{l_s}}{\partial \bar{x}^{j_s}} T^{k_1 k_2 \ldots k_r}_{l_1 l_2 \ldots l_s}$$

11. Fundamental Operations with Tensors

Addition and Subtraction

The sum or difference of two tensors of the same rank and same algebraic type (having identical numbers of contravariant and covariant indices) yields another tensor of the exact same rank and type.

Let A^{ij} and B^{ij} be two contravariant tensors of rank 2. Define their sum as:

$$C^{ij} = A^{ij} + B^{ij}$$

To prove that C^{ij} is a contravariant tensor of rank 2, write the transformation equations for A^{ij} and B^{ij}:

$$\bar{A}^{ij} = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial \bar{x}^j}{\partial x^l}A^{kl}$$

$$\bar{B}^{ij} = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial \bar{x}^j}{\partial x^l}B^{kl}$$

Adding these two transformation equations gives:

$$\bar{A}^{ij} + \bar{B}^{ij} = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial \bar{x}^j}{\partial x^l}\left(A^{kl} + B^{kl}\right)$$

Substituting the definition of C^{ij} yields:

$$\bar{C}^{ij} = \frac{\partial \bar{x}^i}{\partial x^k}\frac{\partial \bar{x}^j}{\partial x^l}C^{kl}$$

Thus, C^{ij} transforms as a contravariant tensor of rank 2.

Outer Product

The outer product (or tensor product) of two tensors of arbitrary ranks produces a new tensor whose rank is the sum of the ranks of the original tensors.

Consider a contravariant vector A^i (rank 1) and a covariant vector B_j (rank 1). Define their outer product as:

$$T^i_j = A^i B_j$$

To establish its transformation law, apply the individual transformation rules for A^i and B_j:

$$\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^k} A^k$$

$$\bar{B}_j = \frac{\partial x^l}{\partial \bar{x}^j} B_l$$

Multiplying these two expressions together yields:

$$\bar{A}^i \bar{B}_j = \left(\frac{\partial \bar{x}^i}{\partial x^k} A^k \right) \left(\frac{\partial x^l}{\partial \bar{x}^j} B_l \right)$$

Rearranging terms gives:

$$\bar{T}^i_j = \frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^j} A^k B_l$$

Substituting $T^k_l = A^k B_l$ results in:

$$\bar{T}^i_j = \frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^j} T^k_l$$

This proves that the outer product T^i_j is a mixed tensor of rank 2.

Contraction

Contraction is the process of equating one contravariant (upper) index and one covariant (lower) index of a mixed tensor and summing over that shared index. This operation reduces the contravariant rank by 1, the covariant rank by 1, and the overall rank of the tensor by 2.

Consider a mixed rank-2 tensor T^i_j with transformation rule:

$$\bar{T}^i_j = \frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^j} T^k_l$$

Equating index j to index i results in the contracted quantity $\bar{T}^i_i$:

$$\bar{T}^i_i = \frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^i} T^k_l$$

Combining the partial derivative factors using the chain rule yields:

$$\frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^i} = \frac{\partial x^l}{\partial x^k} = \delta^l_k$$

where δ^l_k is the Kronecker delta. Substituting this back gives:

$$\bar{T}^i_i = \delta^l_k T^k_l$$

Evaluating the sum over index l using the property of the Kronecker delta:

$$\bar{T}_i^i = T_k^k$$

Because $\bar{T}_i^i = T_k^k$, the contracted component T_i^i remains unchanged under coordinate transformations and is therefore a scalar invariant (tensor of rank 0).

Inner Product

An inner product is formed by taking the outer product of two tensors and subsequently contracting one upper index with one lower index. The resulting tensor has a rank equal to the sum of the ranks of the original tensors minus 2.

Consider a contravariant vector A^i and a covariant vector B_j. The outer product is $T_j^i = A^i B_j$. Contracting the indices by setting $j = i$ gives the inner product:

$$S = A^i B_i$$

To verify that S is a scalar, write the components in the barred coordinate system:

$$\bar{S} = \bar{A}^i \bar{B}_i$$

Substitute the transformation rules for $\bar{A}^i$ and $\bar{B}_i$:

$$\bar{S} = \left(\frac{\partial \bar{x}^i}{\partial x^k} A^k \right) \left(\frac{\partial x^l}{\partial \bar{x}^i} B_l \right)$$

Rearranging terms yields:

$$\bar{S} = \left(\frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^i} \right) A^k B_l$$

Simplifying the product of derivatives using the Kronecker delta:

$$\bar{S} = \delta_k^l A^k B_l$$

Summing over index l:

$$\bar{S} = A^k B_k = S$$

Hence, the inner product of a contravariant vector and a covariant vector is a scalar invariant.

Raising and Lowering of Indices

Indices of tensors are lowered or raised by taking inner products with the metric tensor g_{ij} or its inverse g^{ij}, respectively.

To lower the contravariant index of a vector A^j to form a covariant vector A_i, contract it with the metric tensor g_{ij}:

$$A_i = g_{ij} A^j$$

To raise the covariant index of a vector B_j to form a contravariant vector B^i, contract it with the inverse metric tensor g^{ij}:

$$B^i = g^{ij} B_j$$

For a rank-2 contravariant tensor T^{jk}, lowering both indices is accomplished by applying the metric tensor twice:

$$T_{im} = g_{ij} g_{km} T^{jk}$$

Lowering a single index yields a mixed tensor:

$$T_i^{\ k} = g_{ij} T^{jk}$$

12. Metric Tensor, Covariant Differentiation and Christoffel Symbols

The Metric Tensor

In a general curvilinear coordinate system, the invariant distance squared ds^2 between two infinitely close points x^i and $x^i + dx^i$ is defined by the quadratic differential form:

$$ds^2 = g_{ij} dx^i dx^j$$

where g_{ij} are the components of the metric tensor.

Because $dx^i dx^j = dx^j dx^i$, any antisymmetric part of g_{ij} cancels out in the sum. Thus, the metric tensor is symmetric:

$$g_{ij} = g_{ji}$$

The metric tensor g_{ij} is a covariant tensor of rank 2. Its matrix inverse g^{ij} is defined by the relation:

$$g_{ik} g^{kj} = \delta_i^j$$

where g^{ij} is a contravariant tensor of rank 2, known as the inverse (or conjugate) metric tensor.

Metric Tensor in Spherical Polar Coordinates

Consider 3-dimensional Euclidean space in spherical polar coordinates $(x^1, x^2, x^3) = (r, \theta, \phi)$. The transformation to Cartesian coordinates (x, y, z) is:

$$x = r \sin\theta \cos\phi$$

$$y = r \sin\theta \sin\phi$$

$$z = r\cos\theta$$

The differential spatial displacement in Cartesian coordinates gives the invariant line element:

$$ds^2 = dx^2 + dy^2 + dz^2$$

Taking total differentials of x, y, and z:

$$dx = \sin\theta\cos\phi\, dr + r\cos\theta\cos\phi\, d\theta - r\sin\theta\sin\phi\, d\phi$$

$$dy = \sin\theta\sin\phi\, dr + r\cos\theta\sin\phi\, d\theta + r\sin\theta\cos\phi\, d\phi$$

$$dz = \cos\theta\, dr - r\sin\theta\, d\theta$$

Squaring each differential and summing $dx^2 + dy^2 + dz^2$ yields:

$$ds^2 = (dr)^2 + r^2(d\theta)^2 + r^2\sin^2\theta(d\phi)^2$$

Comparing this expression to the general metric form $ds^2 = g_{ij}dx^i dx^j$ with $(dx^1, dx^2, dx^3) = (dr, d\theta, d\phi)$, the components of the metric tensor are:

$$g_{11} = 1$$

$$g_{22} = r^2$$

$$g_{33} = r^2\sin^2\theta$$

and off-diagonal components $g_{ij} = 0$ for $i \neq j$. Written in matrix form:

$$[g_{ij}] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{pmatrix}$$

Failure of Ordinary Differentiation

The ordinary partial derivative of a tensor component with respect to a coordinate does not generally transform as a tensor.

To demonstrate this failure, consider a contravariant vector field $A^i(x)$ with transformation law:

$$\bar{A}^i = \frac{\partial \bar{x}^i}{\partial x^k} A^k$$

Differentiating both sides with respect to $\bar{x}^j$:

$$\frac{\partial \bar{A}^i}{\partial \bar{x}^j} = \frac{\partial}{\partial \bar{x}^j}\left(\frac{\partial \bar{x}^i}{\partial x^k} A^k\right)$$

Applying the chain rule $\frac{\partial}{\partial \bar{x}^j} = \frac{\partial x^l}{\partial \bar{x}^j}\frac{\partial}{\partial x^l}$ to the right-hand side:

$$\frac{\partial \bar{A}^i}{\partial \bar{x}^j} = \frac{\partial x^l}{\partial \bar{x}^j} \frac{\partial}{\partial x^l} \left(\frac{\partial \bar{x}^i}{\partial x^k} A^k \right)$$

Expanding using the product rule:

$$\frac{\partial \bar{A}^i}{\partial \bar{x}^j} = \frac{\partial \bar{x}^i}{\partial x^k} \frac{\partial x^l}{\partial \bar{x}^j} \frac{\partial A^k}{\partial x^l} + A^k \frac{\partial^2 \bar{x}^i}{\partial x^l \partial x^k} \frac{\partial x^l}{\partial \bar{x}^j}$$

The presence of the second term involving second derivatives $\frac{\partial^2 \bar{x}^i}{\partial x^l \partial x^k}$ breaks the tensorial transformation law. Thus, the ordinary derivative $\frac{\partial A^i}{\partial x^j}$ is not a tensor unless the transformation coefficients are constant everywhere (as in Cartesian systems).

Christoffel Symbols

To construct a differential operator that preserves tensorial properties, connection coefficients called Christoffel symbols are introduced to compensate for the variation of spatial basis vectors.

Christoffel Symbol of the First Kind

The Christoffel symbol of the first kind, denoted $[ij, k]$ or Γ_{ijk}, is defined in terms of metric tensor derivatives as:

$$[ij, k] = \frac{1}{2} \left(\frac{\partial g_{ik}}{\partial x^j} + \frac{\partial g_{jk}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^k} \right)$$

It is symmetric with respect to its first two indices:

$$[ij, k] = [ji, k]$$

Christoffel Symbol of the Second Kind

The Christoffel symbol of the second kind, denoted Γ^k_{ij}, is obtained by raising the third index of the Christoffel symbol of the first kind using the inverse metric tensor:

$$\Gamma^k_{ij} = g^{kl} [ij, l]$$

Explicitly in terms of the metric tensor components:

$$\Gamma^k_{ij} = \frac{1}{2} g^{kl} \left(\frac{\partial g_{il}}{\partial x^j} + \frac{\partial g_{jl}}{\partial x^i} - \frac{\partial g_{ij}}{\partial x^l} \right)$$

It is also symmetric in its lower indices:

$$\Gamma^k_{ij} = \Gamma^k_{ji}$$

Transformation Law of Christoffel Symbols

Under a coordinate transformation $x \to \bar{x}$, the Christoffel symbol of the second kind transforms according to:

$$\bar{\Gamma}^i_{jk} = \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}\frac{\partial x^p}{\partial \bar{x}^k}\Gamma^m_{np} + \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial^2 x^m}{\partial \bar{x}^j \partial \bar{x}^k}$$

Because of the second non-tensorial term $\frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial^2 x^m}{\partial \bar{x}^j \partial \bar{x}^k}$, Christoffel symbols are not tensors.

Covariant Differentiation

By combining the ordinary partial derivative with a correction term involving Christoffel symbols, an operator called the covariant derivative is defined, which transforms correctly as a tensor.

Covariant Derivative of a Contravariant Vector

The covariant derivative of a contravariant vector field A^i with respect to x^j is denoted by $\nabla_j A^i$ or $A^i{}_{;j}$ and is defined as:

$$A^i{}_{;j} = \nabla_j A^i = \frac{\partial A^i}{\partial x^j} + \Gamma^i_{jk} A^k$$

To verify that $A^i{}_{;j}$ transforms as a mixed rank-2 tensor, evaluate $\bar{A}^i{}_{;\bar{j}}$ in the barred coordinate frame:

$$\bar{A}^i{}_{;\bar{j}} = \frac{\partial \bar{A}^i}{\partial \bar{x}^j} + \bar{\Gamma}^i_{jk}\bar{A}^k$$

Substituting the expressions for $\frac{\partial \bar{A}^i}{\partial \bar{x}^j}$, $\bar{\Gamma}^i_{jk}$, and $\bar{A}^k$:

$$\frac{\partial \bar{A}^i}{\partial \bar{x}^j} = \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}\frac{\partial A^m}{\partial x^n} + A^p\frac{\partial^2 \bar{x}^i}{\partial x^n \partial x^p}\frac{\partial x^n}{\partial \bar{x}^j}$$

$$\bar{\Gamma}^i_{jk}\bar{A}^k = \left(\frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}\frac{\partial x^p}{\partial \bar{x}^k}\Gamma^m_{np} + \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial^2 x^m}{\partial \bar{x}^j \partial \bar{x}^k}\right)\left(\frac{\partial \bar{x}^k}{\partial x^q}A^q\right)$$

Simplifying the second term of the Christoffel contraction using $\frac{\partial x^p}{\partial \bar{x}^k}\frac{\partial \bar{x}^k}{\partial x^q} = \delta^p_q$:

$$\bar{\Gamma}^i_{jk}\bar{A}^k = \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}\Gamma^m_{nq}A^q + A^q\frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial^2 x^m}{\partial \bar{x}^j \partial \bar{x}^k}\frac{\partial \bar{x}^k}{\partial x^q}$$

Using the identity for inverse matrix derivatives:

$$A^q\frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial^2 x^m}{\partial \bar{x}^j \partial \bar{x}^k}\frac{\partial \bar{x}^k}{\partial x^q} = -A^p\frac{\partial^2 \bar{x}^i}{\partial x^n \partial x^p}\frac{\partial x^n}{\partial \bar{x}^j}$$

Adding $\frac{\partial \bar{A}^i}{\partial \bar{x}^j}$ and $\bar{\Gamma}^i_{jk}\bar{A}^k$ causes the non-tensorial second derivatives to cancel completely, leaving:

$$\bar{A}^i{}_{;\bar{j}} = \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}\left(\frac{\partial A^m}{\partial x^n} + \Gamma^m_{nq}A^q\right)$$

Substituting the definition $A^m{}_{;n} = \frac{\partial A^m}{\partial x^n} + \Gamma^m_{nq}A^q$ yields:

$$\bar{A}^i{}_{;\bar{j}} = \frac{\partial \bar{x}^i}{\partial x^m}\frac{\partial x^n}{\partial \bar{x}^j}A^m{}_{;n}$$

This confirms that $A^{i}_{;j}$ transforms as a mixed tensor of rank 2.

Covariant Derivative of a Covariant Vector

The covariant derivative of a covariant vector field A_i with respect to x^j is denoted by $\nabla_j A_i$ or $A_{i;j}$ and is defined as:

$$A_{i;j} = \nabla_j A_i = \frac{\partial A_i}{\partial x^j} - \Gamma^k_{ij} A_k$$

This quantity transforms as a covariant tensor of rank 2 under arbitrary coordinate transformations.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **There is no Assam University question for this unit in the paper bank** — gap. The questions below are topic-matched from other universities' papers (non-AU) or standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] Define a random variable and distinguish discrete and continuous random variables.

Answer in brief: Section 1 — a random variable assigns a number to each outcome of an experiment; discrete takes isolated values (probability mass function), continuous takes values in an interval (probability density function).

S2. [standard] Write the probability mass function of the binomial distribution and state its mean and variance.

Answer in brief: Section 1 — $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$; mean np, variance $np(1-p)$.

S3. [standard] Write the Poisson distribution and state when it is used as an approximation to the binomial.

Answer in brief: Section 2 — $P(X = k) = e^{-\lambda}\lambda^k / k!$; used when n is large, p is small and $np = \lambda$ is moderate.

S4. [standard] State the mean and variance of the Poisson distribution and one property of the normal distribution.

Answer in brief: Sections 2–3 — for Poisson, mean = variance = λ; the normal distribution is symmetric about its mean, with about 68% of the probability within one standard deviation.

S5. [standard] State the Central Limit Theorem.

Answer in brief: Section 4 — the sum (or mean) of a large number of independent, identically distributed random variables is approximately normally distributed, whatever the original distribution.

S6. [standard] State the Law of Large Numbers.

Answer in brief: Section 4 — as the number of trials grows, the sample mean converges (in probability) to the expectation value.

S7. [standard] What is hypothesis testing? Define null and alternative hypotheses.

Answer in brief: Section 5 — a procedure to decide, from data, whether to reject a stated hypothesis; the null hypothesis H_0 is the claim tested, the alternative H_1 is what is accepted if H_0 is rejected.

S8. [standard] Distinguish systematic and random errors, with one example of each.

Answer in brief: Section 6 — systematic errors shift all measurements the same way (e.g. a miscalibrated scale) and are not reduced by repetition; random errors scatter about the true value (e.g. reading fluctuation) and are reduced by averaging.

S9. [non-AU] The scale factors corresponding to the covariant metric tensor g_ij in spherical polar coordinates are — (A) 1, r^2, r^2 sin^2 theta (B) 1, r, r sin theta (C) 1, r^2, r^2 (D) 1, 1, 1. — GATE PH 2018, Q5.

Answer in brief: Section 12 — (A) $1, r^2, r^2 \sin^2\theta$, from the line element $ds^2 = dr^2 + r^2 d\theta^2 + r^2 \sin^2\theta \, d\phi^2$.

S10. [standard] Define the metric tensor and the Christoffel symbols.

Answer in brief: Section 12 — the metric tensor g_{ij} gives the line element $ds^2 = g_{ij} dx^i dx^j$ and raises/lowers indices; the Christoffel symbols Γ^k_{ij} are built from derivatives of g_{ij} and appear in the covariant derivative.

Long-form questions

L1. [non-AU] Find the probability of almost 5 defective fuses in a box of 200 fuses, if 2% are defective. — Gauhati University, B.Sc 4th Sem Physics M-4.1, 2015, Q2c, 5 marks (binomial/Poisson).

What to write: Section 2 in full — set up $n = 200$, $p = 0.02$, $\lambda = np = 4$; use the Poisson distribution to find $P(X \le 5)$ (or $P(X = 5)$), summing $e^{-4}4^k/k!$ as needed.

L2. [non-AU] What is Gaussian distribution? Define standard deviation. Prove the theory of compound probability. State one property of normal distribution. — Gauhati University, B.Sc 4th Sem Physics M-4.1, 2015, Q5c.

What to write: Section 3 in full — the normal density $\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$, standard deviation as the square root of the variance, the compound (product) rule for independent probabilities, and a property such as symmetry about the mean or the 68-95-99.7 rule.

L3. [standard] Explain systematic and random errors, derive the propagation of errors for a sum and a product, define standard and probable error, and fit a straight line to given data by the least-squares method.

What to write: Sections 6–9 in full — error types with examples; for $z = x + y$, $\Delta z = \sqrt{(\Delta x)^2 + (\Delta y)^2}$; for $z = xy$, $\Delta z/z = \sqrt{(\Delta x/x)^2 + (\Delta y/y)^2}$; standard error $\sigma/\sqrt{n}$, probable error 0.6745σ; least-squares normal equations for $y = a + bx$.

L4. [standard] Define covariant and contravariant tensors, explain the metric tensor, and derive the Christoffel symbols and the covariant derivative.

What to write: Sections 10–12 in full — transformation laws for upper and lower indices, raising and lowering with g_{ij}, the metric in spherical polar coordinates, and $\Gamma^k_{ij} = \frac{1}{2} g^{kl}(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$ with the covariant derivative $\nabla_j A^i = \partial_j A^i + \Gamma^i_{jk} A^k$.

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.